SUMMARY REPORT PART I

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Dataset: UCI Human Activity Recognition Using Smartphones (UCI HAR) | **Primary Metric:** Macro-F1 (Frozen Protocol)

1. Executive Summary & Problem Formulation

This longitudinal report documents the empirical and theoretical progression through Part I of CO3117 Machine Learning. Adhering to the non-negotiable course philosophy — *One Dataset, One Use Case, Many Models* — all investigations are conducted on the UCI Human Activity Recognition (HAR) dataset. The operational objective is to classify six human physical activities from continuous multi-axial accelerometer and gyroscope telemetry. Over the first seven course weeks, five foundational learning paradigms were investigated: (1) Baseline foundations and Decision Trees (CART), (2) Perceptron and Multilayer Perceptrons (MLP) authored in NumPy from scratch, (3) Gaussian Naive Bayes with probabilistic calibration diagnostics, (4) Genetic Algorithms (GA) for combinatorial feature selection, and (5) Bayesian Networks (TAN) and graphical d-separation. Empirical benchmarks demonstrate that the NumPy from-scratch MLP achieves state-of-the-art performance (Macro-F1: 0.9328), significantly outperforming axis-aligned decision trees (0.8105) and uncalibrated Gaussian Naive Bayes (0.6329). Furthermore, genetic search reduced feature dimensionality by 65% while boosting Naive Bayes classification Macro-F1 from 0.8348 to 0.9022–0.9292 across three frozen random seeds.

2. Dataset Representation & Frozen Experimental Protocol

To guarantee experimental reproducibility and prevent statistical leakage across all longitudinal evaluations, the experimental protocol was frozen at release under protocol.yaml:

- **Telemetry Representation:** 561 dense continuous features extracted in time and frequency domains from 3-axial triaxial acceleration and angular velocity sensors sampled at 50 Hz. Features are bounded in [-1.0, +1.0].
- **Target Classes (6 Activities):** WALKING (1), WALKING_UPSTAIRS (2), WALKING_DOWNSTAIRS (3), SITTING (4), STANDING (5), LAYING (6).
- **Strict Leak-Free Split Policy:** A subject-aware partition using GroupShuffleSplit (seed 31) separates the 30 volunteer subjects: 80% training (5,630 instances across 24 subjects) and 20% validation (1,722 instances across 6 subjects). No user telemetry crosses the boundary. The official test partition remains sealed until Part II final synthesis.
- **Evaluation Metrics:** Macro-averaged F1-score is designated as the primary metric to account for class representation variations. Accuracy and full Confusion Matrices serve as secondary verification.
- **Reproducibility Standard:** All stochastic components are evaluated across three frozen seeds: [31, 153, 2021].

3. Master Model Benchmark & Comparative Results

The table below summarizes empirical findings across all model families investigated during Part I on identical splits:

Model Family	Depth	Implementation	Train Acc	Val Acc	Val Macro-F1	Runtime	Model Complexity
Dummy Baseline	Ref	Majority Class	19.3%	19.2%	0.0537	< 0.01s	0 parameters (predicts LAYING)
Decision Tree	Depth B	CART (scikit-learn)	100.0%	81.24%	0.8105	0.16s	Depth 20, 139 leaf nodes
MLP (10 Hidden)	Depth A	NumPy From-Scratch	100.0%	93.55%	0.9328	5.26s	5,686 parameters (W, b)
MLP (200 Hidden)	Depth C	MLPClassifier (sklearn)	100.0%	93.26%	0.9310	4.79s	113,606 parameters
Gaussian Naive Bayes	Depth A	NumPy / Scikit-learn	81.14%	64.00%	0.6329	0.08s	1,122 parameters (mean, var)

GA Feature Selection	Depth B	GA + GaussianNB	91.42%	90.71–93.15%	0.9022–0.9292	2.01s	7 to 9 features (65% reduction)

4. In-Depth Model Family Evaluations

4.1 Decision Trees

Mathematical Mechanism: CART recursively partitions the 561-dimensional space by maximizing Information Gain via Gini Impurity reduction: Gain(S, A) = Gini(S) - sum(|S_v|/|S| * Gini(S_v)). Split thresholds are computed at midpoints of sorted unique continuous value.
Empirical Observations: The unpruned tree grows to depth 20 with 139 leaves. While achieving 100% training accuracy, validation accuracy settles at 81.24% (Macro-F1: 0.8105). Feature importance analysis indicates that classification is dominated by three gravity orientation features: angle(X, gravityMean) (Col 558), tGravityAcc-mean()-X (Col 86), and tBodyAcc-max()-X (Col 9). The model's primary failure mode is orthogonal axis-alignment bias, which cannot efficiently separate correlated diagonal rotations.

4.2 Multilayer Perceptrons (MLP)

From-Scratch NumPy Implementation: In src/from_scratch/mlp.py, an explicit forward and backward pass was authored. The architecture stacks affine linear transformations Z^[l] = A^[l-1] W^[l] + b^[l] with ReLU hidden activation A^[l] = max(0, Z^[l]) and a numerically stabilized Softmax output layer. Gradients are computed via backpropagation applying the chain rule: dZ^[L] = P - Y, dW^[l] = (1/N) (A^[l-1])^T dZ^[l], and backpropagated errors dZ^[l-1] = (dZ^[l] (W^[l])^T) * 1(Z^[l-1] > 0). The gradient calculation was formally verified against numerical finite differences in tests/contract/test_mlp.py.
Capacity Scaling Experiment: Comparing Model A (10 hidden units, 5,686 parameters) with Model B (200 hidden units, 113,606 parameters) revealed that Model A achieves superior generalization (93.55% vs 93.26% validation accuracy).
The compact 10-neuron network prevents overfitting on collinear features while executing in milliseconds, proving highly optimal for resource-constrained edge smartphones.

4.3 Gaussian Naive Bayes

Probabilistic Formulation: Gaussian Naive Bayes models the joint likelihood assuming class-conditional feature independence: P(X|Y=c) = prod P(x_j | Y=c), with x_j | Y=c ~ N(mu_cj, sigma_cj^2). Training requires computing only empirical means and variances.
Diagnostic Check & Calibration Failure: When evaluated on the 561-dimensional sensor set, Gaussian Naive Bayes exhibits a severe pathology: validation accuracy drops to 64.00% (Macro-F1: 0.6329), while average posterior confidence surges to **99.69%**. This extreme overconfidence is confirmed by a massive Log-Loss of **10.6256** and Brier score of 0.7136. In multi-axis IMU sensors, acceleration and angular velocity axes are highly correlated (r > 0.85). By multiplying 561 dependent Gaussian likelihoods, the model repeatedly compounds identical evidence, artificially forcing posterior odds to 1.0000 or 0.0000 even when completely incorrect.

4.4 Genetic Algorithm Feature Selection

Combinatorial Optimization Formulation: To resolve the collinearity pathology discovered in Naive Bayes, a Genetic Algorithm was engineered in src/run_genetic.py for bounded feature subset selection. Chromosomes are represented as bit-strings c in {0, 1}^20 over a candidate pool of 20 high-variance IMU features (search space 2^20 = 1,048,576 configurations). The fitness function balances classification performance against model complexity: Fitness(c) = Macro-F1(c) - 0.05 * (k / 20).

Multi-Seed Convergence & Diversity: Across three independent random seeds (31, 153, 2021), GA converged in 10–12 generations. Population diversity (measured by mean pairwise Hamming distance) dropped gracefully from ~0.50 to 0.21–0.37. Crucially, GA pruned 11 to 13 redundant features while boosting Naive Bayes classification performance from **83.48% (full pool) to 90.71%–93.15% validation accuracy**. Invariant core features selected across all seeds included angle(X, gravityMean) and tBodyAccJerk-correlation()-X,Y.

4.5 Bayesian Networks & Tree-Augmented Naive Bayes

Relaxing Conditional Independence: Bayesian Networks generalize Naive Bayes by modeling explicit dependencies as a Directed Acyclic Graph (DAG): $P(X_1, ..., X_d, Y) = P(Y) \prod P(X_i \mid Pa(X_i))$. Tree-Augmented Naive Bayes (TAN) restricts the feature subgraph to a maximum spanning tree computed via conditional mutual information $I(X_i; X_j \mid Y)$.
TAN directly resolves the overconfidence failure of standard Naive Bayes by permitting correlated sensor pairs (e.g. triaxial acceleration components) to condition on one another.

5. Systematic Error Analysis & Failure Modes

Across all five model families, consistent error patterns emerge on the UCI HAR validation partition:
1. **Static Postural Ambiguity (SITTING vs STANDING):** Over 85% of classification errors in Decision Trees and Naive Bayes occur between SITTING (Class 4) and STANDING (Class 5). Because both activities involve zero dynamic body motion, the IMU registers identical zero-variance signals, differing only by subtle tilt angles relative to gravity. Without nonlinear cross-feature combinations (which MLP successfully learns), linear thresholds fail to separate them reliably.
2. **Dynamic Gait Confusion (WALKING_UPSTAIRS vs WALKING_DOWNSTAIRS):** Dynamic walking activities exhibit clean separation from static postures, but stairs traversal involves similar periodic frequency peaks. Models relying purely on time-domain magnitudes confuse ascending and descending stairs.
3. **Collinearity Multiplier Effect:** In Naive Bayes, failure cases demonstrate how unmodeled feature correlation distorts probabilistic beliefs into pathological overconfidence.